

Complete Flexbox and CSS Grid Notes

These notes cover all important concepts of CSS Flexbox and CSS Grid including properties, syntax, examples, responsive design, and project ideas.

1. CSS Flexbox

Flexbox is a one-dimensional layout system used to arrange items in rows or columns. It helps developers build responsive layouts easily.

Basic Syntax

```
.container { display: flex; }
```

Property	Description
display:flex	Makes the container flexible
flex-direction	Sets row or column direction
justify-content	Aligns items horizontally
align-items	Aligns items vertically
flex-wrap	Allows items to wrap
gap	Adds spacing between items
align-content	Controls wrapped row spacing

Important Flexbox Concepts

- flex-direction: row, row-reverse, column, column-reverse
- justify-content: center, space-between, space-around, space-evenly
- align-items: center, flex-start, flex-end, stretch
- flex-wrap: wrap or nowrap
- order changes the position of items
- flex-grow expands item size
- flex-shrink reduces item size
- flex-basis sets initial item width

Flexbox Example

```
.container { display: flex; justify-content: center; align-items: center; gap: 20px; }
```

2. CSS Grid

CSS Grid is a two-dimensional layout system that works with rows and columns. It is mainly used for complete webpage layouts.

Basic Syntax

```
.container { display: grid; grid-template-columns: repeat(3, 1fr); }
```

Property	Description
display:grid	Creates grid container
grid-template-columns	Defines column sizes
grid-template-rows	Defines row sizes
gap	Adds spacing between rows and columns
grid-column	Controls column span
grid-row	Controls row span
place-items	Aligns items easily
justify-items	Horizontal alignment
align-items	Vertical alignment

Important Grid Concepts

- 1fr means one fraction of available space
- repeat() repeats rows or columns
- minmax() sets minimum and maximum size
- auto-fit adjusts columns automatically
- auto-fill creates automatic tracks
- grid-column: span 2 takes two columns
- grid-row: span 2 takes two rows

Grid Example

```
.container { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px; }
```

3. Flexbox vs Grid

Flexbox	Grid
One-dimensional layout	Two-dimensional layout
Works in row OR column	Works in rows AND columns
Best for alignment	Best for full layouts
Simple layouts	Complex layouts

4. Responsive Design using Media Query

```
@media(max-width:768px) { .container { grid-template-columns: 1fr; } }
```

5. Flex and Grid Project Ideas

- Portfolio Website
- Blog Website
- Photo Gallery
- Admin Dashboard
- E-commerce Website
- Restaurant Website
- Task Management App
- Responsive Navbar

6. Interview Questions

- What is Flexbox?
- What is CSS Grid?
- Difference between Flexbox and Grid?
- What is justify-content?
- What is align-items?
- What is 1fr in Grid?
- Difference between auto-fit and auto-fill?
- How to make responsive layouts?